

Course Assignment (Project 130): Integrating Cancer Mutations, Gene Expression, and Neoantigen Prediction

Part 0: Project Orientation

1. Project Overview

Students will work in groups of two (groups of three are permitted). Each group will select one human cancer type and construct an integrated dataset containing:

1. Somatic mutations across multiple tumor samples
2. Mutation occurrence per tumor sample
3. Gene-expression levels measured as transcripts per million (TPM)

For the advanced component, students will translate coding mutations into mutant peptides and evaluate their neoantigen potential.

Each group should analyze a different cancer type unless the instructor grants permission.

2. Learning Objectives

By completing this project, students should be able to:

1. Download and process public cancer genomics data
2. Interpret somatic mutation annotation files
3. Construct mutation-by-sample and gene-by-sample matrices
4. Process and integrate RNA-seq expression data
5. Map genomic variants to coding and protein consequences
6. Generate mutation-containing peptides
7. Apply peptide-MHC binding and immunogenicity prediction tools
8. Produce a reproducible bioinformatics workflow

Part I. Core Requirements

3. Selection of a Cancer Type

Select one cancer type, such as:

1. Breast cancer
2. Colorectal cancer
3. Osteosarcoma
4. Gastric cancer
5. Biliary tract neoplasms
6. Hepatocellular carcinoma
7. Lung cancer

8. Pancreatic cancer
9. Bladder cancer
10. Ovarian cancer

(Note: For these example cancers, we can get first hand hospital data for validation if needed as future extension/publication of this project, but it's not restricted to these cancer types, you can choose the ones you are most interested in.)

Each group should preferably include:

1. At least 20 tumor samples in the mutation dataset;
2. At least 10 tumor samples in the gene-expression dataset.

Smaller datasets may be accepted when the selected cancer is rare, but the reason must be explained.

4. Somatic Mutation Dataset

Obtain somatic mutation data from a public cancer genomics resource, preferably The Cancer Genome Atlas through the NCI Genomic Data Commons.

GDC project-level Mutation Annotation Format files are tab-delimited files containing aggregated somatic mutation information.

The mutation table should contain, where available:

- Gene symbol;
- Chromosome;
- Genomic position;
- Reference allele;
- Alternative allele;
- Variant classification;
- Transcript identifier;
- Coding DNA change;
- Protein or amino-acid change;
- Tumour sample identifier.

For the core assignment, students may restrict the analysis to:

- Protein-coding genes;
- PASS or otherwise high-confidence variants;
- Nonsynonymous somatic mutations;
- Missense single-nucleotide variants.

Students must document the reference genome assembly used, for example:

- GRCh37/hg19; or
- GRCh38/hg38.

Do not mix genomic coordinates from different reference genome assemblies.

5. Mutation-by-Sample Matrix

Construct a binary mutation-by-sample matrix. Each row represents a specific mutation; each column represents a tumour sample. Use 1 if the mutation is present, 0 if absent.

Example (tab-delimited):

Gene_Name	Mutation	AminoAcid_Change	Sample1	Sample2	Sample3
ARID1A	c.296C>G	p.Pro99Arg	1	0	1
TP53	c.743G>A	p.Arg248Gln	0	1	1
KRAS	c.35G>A	p.Gly12Asp	1	0	0

Use HGVS notation (e.g., c.35G>A, p.Gly12Asp). Represent multiple mutations in the same gene as separate rows. If multiple mutations occur in the same gene, they should normally be represented as separate rows.

6. Gene-Expression Dataset

Obtain an RNA-seq dataset for the same cancer type from GEO or another approved repository. Select RNA-seq data, not microarray.

Construct a gene-by-sample TPM matrix, for example:

GeneName	ExprSample1	ExprSample2	ExprSample3
ARID1A	18.6	21.4	20.1
TP53	42.1	38.7	45.2
KRAS	31.5	29.8	34.6

Students may use:

1. TPM values supplied by the original study; or
2. Raw RNA-sequencing counts converted to TPM.

The following information must be reported:

- GEO accession number
- Sequencing platform
- Number of tumour and normal samples

- Whether TPM was downloaded or calculated
- Gene identifier type
- Method used to map identifiers to gene symbols
- Method used to handle duplicated gene symbols

Values reported as read counts, FPKM or RPKM must not be labelled as TPM without appropriate conversion.

7. Integration of Mutations and Gene Expression

Since mutation samples (TCGA) and expression samples (TCGA or GEO) generally differ, calculate a cancer-level expression summary:

$$GeneLevelTPM = median\ TPM\ across\ tumour\ samples$$

The integrated matrix should be tab-delimited:

GeneName	Mutation	AminoAcidChange	GeneLevelTPM	Sample1	Sample2
ARID1A	c.296C>G	p.Pro99Arg	20.1	1	0
TP53	c.743G>A	p.Arg248Gln	42.1	0	1
KRAS	c.35G>A	p.Gly12Asp	31.5	1	0

State the aggregation method. Median TPM is recommended as it is robust to extreme values.

Alternatively, groups may use matched TCGA mutation and RNA-seq data from the same samples; the matching procedure must be clearly explained.

8. Basic Data Quality Control

Students should perform and report at least the following checks:

- Number of mutations before and after filtering;
- Number of unique genes
- Number of tumour samples
- Percentage of missing expression values
- Consistency of gene identifiers
- Consistency of reference genome assembly
- Distribution of GeneLevelTPM values
- The ten most frequently mutated genes
- The ten most highly expressed mutated genes

At least two figures should be included, for example:

1. Bar plot of the most frequently mutated genes

2. Histogram or density plot of GeneLevelTPM
3. Heat map of selected mutations across samples
4. Plot comparing mutation frequency with gene expression

Part II. Advanced Requirements: Mutant Peptides and Neoantigens

9. Variant-to-Protein Annotation

Map each selected coding mutation to its affected transcript and protein sequence.

Ensembl Variant Effect Predictor may be used to determine transcript-level and protein-level consequences. VEP accepts genomic variants and predicts their effects on transcripts and protein sequences, and a dedicated GRCh37 version is available for analyses using hg19/GRCh37 coordinates.

For each mutation, report:

- Reference genome assembly;
- Gene symbol;
- Transcript ID;
- Protein ID, when available;
- Genomic variant;
- Coding DNA change;
- Protein change;
- Variant consequence;
- Protein position;
- Transcript-selection rule.

A consistent transcript-selection strategy must be used, such as:

- MANE Select transcript;
- Ensembl canonical transcript;
- Most highly expressed transcript;
- A clearly justified biologically relevant transcript.

For the main advanced analysis, students may initially restrict the analysis to missense mutations. In-frame insertions/deletions and frameshift mutations may be included as an additional extension.

10. Mutant Peptide Generation

For every eligible missense mutation, generate:

- All mutation-containing 9-mer peptides;
- All mutation-containing 15-mer peptides;

- Corresponding wild-type peptides.

Every reported peptide must contain the altered amino acid.

For a mutation at protein position (m), generate all peptide windows in which position (m) is included, provided that the window remains within the protein sequence.

Report the mutation position within each peptide using **1-based indexing**.

Example:

GeneName	ProteinChange	Peptide	Length	MutPos	Type
KRAS	p.Gly12Asp	VVGADGVGK	9	5	Mutant
KRAS	p.Gly12Asp	VVGAGGVGK	9	5	WildType

Students must verify that:

- The reference amino acid agrees with the reference protein sequence;
- The mutant amino acid has been inserted at the correct position;
- The peptide does not cross the protein boundary;
- The wild-type and mutant peptides differ only at the mutated position for a missense variant.

11. HLA Selection

Peptide–MHC prediction requires one or more HLA alleles. Students should use either:

Option A (Fixed panel):

At least 3 common HLA class I alleles (e.g., HLA-A*02:01, HLA-A*01:01, HLA-A*03:01) plus at least 2 HLA class II alleles for 15-mer analysis.

Option B (Sample-specific):

Use patient-specific HLA types when reliable HLA-typing results are available. The data source and HLA-typing method must be documented.

Students must not present peptide–MHC binding scores without reporting the corresponding HLA allele.

12. Peptide-MHC Binding Prediction

12.1 Nine-mer peptides

Evaluate mutant and wild-type 9-mer peptides using at least one HLA class I binding or presentation predictor, such as:

- NetMHCpan;
- MHCflurry;
- MixMHCpred;
- An equivalent method approved by the instructor.

NetMHCpan 4.2 predicts peptide binding or ligand presentation for MHC molecules of known sequence. MHCflurry provides command-line and Python interfaces for pan-allele MHC-I binding and presentation prediction.

Record at least:

- HLA allele;
- Predicted binding affinity, if available;
- Percentile rank;
- Binding or presentation score;
- Binder classification;
- Software name;
- Software version;
- Prediction mode.

12.2 Fifteen-mer peptides

Evaluate 15-mer peptides using an HLA class II predictor such as NetMHCIIpan. The current NetMHCIIpan 4.2 service supports prediction of peptide presentation by MHC class II molecules.

Because HLA-I and HLA-II prediction models represent different antigen-presentation pathways, results for 9-mer and 15-mer peptides should be reported separately.

13. Immunogenicity Prediction

For mutation-containing HLA class I peptides, apply at least one immunogenicity prediction method.

One recommended option is PRIME, a class I immunogenicity predictor that combines MHC-I binding with features related to T-cell receptor recognition.

Report:

- Peptide sequence;
- HLA allele;
- Immunogenicity score;
- Score direction, indicating whether a higher or lower value is better;
- Software and version;
- Any prediction threshold used.

Students must distinguish between:

1. **MHC binding or presentation**, meaning that a peptide is predicted to be presented by an HLA molecule; and
2. **Immunogenicity**, meaning that a presented peptide is predicted to have the potential to stimulate a T-cell response.

A strong predicted binder is not automatically immunogenic.

14. Comparison with Wild-Type Peptide

For each mutant peptide, compare with its wild-type counterpart. Calculate:

$$\text{DeltaAffinity} = \text{WildTypeAffinity} - \text{MutantAffinity}$$

or

$$\text{AffinityFoldChange} = \text{WildTypeAffinity} / \text{MutantAffinity}$$

Because a lower predicted IC50 generally represents stronger binding, students must explain the direction of the metric they use.

Priority neoantigen candidates should preferably show some combination of:

- Strong mutant peptide–HLA binding;
- Better mutant binding than wild-type binding;
- Favourable immunogenicity score;
- Expression of the mutated gene;
- Occurrence in one or more tumour samples;
- Low or absent presentation predicted for the wild-type peptide.

These criteria should be treated as prioritisation features rather than proof of experimental immunogenicity.

15. Advanced Output Format

Submit a tab-delimited neoantigen table with columns such as:

GeneName
Chromosome
Position
Ref
Alt
TranscriptID
ProteinChange
GeneLevelTPM
MutationFrequency

PeptideType
 Peptide
 PeptideLength
 MutationPosition
 HLAAllele
 BindingAffinity
 BindingRank
 BindingScore
 ImmunogenicityScore
 PredictionTool
 ToolVersion

Example:

Gene Name	ProteinChange	GeneLevelTPM	PeptideType	Peptide	Length	Mut Pos	HLA Allele	BindAff	Bind Rank	Immun Score
KRAS	p.Gly12 Asp	31.5	Mutant	VVGAD GVGK	9	5	HLA-A*03:01	85.2	0.42	0.73
KRAS	p.Gly12 Asp	31.5	WildType	VVGAG GVGK	9	5	HLA-A*03:01	760.4	3.80	0.18

Represent missing tool outputs as NA, not zero.

Part III. Deliverables

A. Data files

- 01_mutation_by_sample.tsv
- 02_gene_by_sample_TPM.tsv
- 03_integrated_mutation_expression.tsv
- 04_neoantigen_predictions.tsv (advanced component only)

B. Source code

Submit all R, Python or shell scripts required to reproduce the analysis. Scripts should be numbered in the order in which they are executed, for example:

```

01_download_mutations.py
02_build_mutation_matrix.py
03_process_expression.R
04_integrate_datasets.py
05_generate_peptides.py
06_predict_neoantigens.py
  
```

C. README file

The README must include:

- Group members;
- Selected cancer type;
- Dataset accession numbers;
- Download dates;
- Reference genome assembly;
- Software dependencies;
- Software versions;
- Instructions for running the pipeline;
- Explanation of all output columns;
- Known limitations.

D. Brief report

Submit a report of approximately 1,500-2,500 words containing:

1. Background and selected cancer type;
2. Data sources;
3. Methods;
4. Quality-control results;
5. Mutation and expression results;
6. Neoantigen results, when applicable;
7. Biological interpretation;
8. Limitations;
9. Contributions of each group member.

E. Oral presentation

Each group will give a short presentation covering:

1. Dataset selection;
2. Computational workflow;
3. Major findings;
4. One technical challenge;
5. One biological conclusion.

Part IV. Assessment

Core component: 80 points

- **Dataset selection and documentation (10 points)**

- Appropriate cancer type and datasets;
- Accurate metadata;
- Clear reference genome specification.
- **Mutation processing (20 points)**
 - Correct mutation filtering;
 - Correct variant representation;
 - Accurate mutation-by-sample matrix;
 - Appropriate handling of duplicated records.
- **Expression processing (20 points)**
 - Correct TPM matrix;
 - Appropriate gene identifier conversion;
 - Transparent aggregation method;
 - Appropriate treatment of missing values.
- **Data integration (15 points)**
 - Correct matching by gene;
 - Accurate final tab-delimited matrix;
 - Clear distinction between mutation samples and expression samples;
 - Appropriate GeneLevelTPM calculation.
- **Reproducibility and presentation (15 points)**
 - Organised and readable code;
 - README and software versions;
 - Informative figures;
 - Clear interpretation.

Advanced component: 20 points

- **Variant-to-peptide mapping (5 points)**
 - Correct transcript and protein annotation;
 - Correct mutant and wild-type sequences;
 - Correct 9-mer (for MHC-I, targeting CD8+ T cells) and 15-mer (for MHC-II, targeting CD4+ T cells) generation;
 - Correct mutation position.
- **MHC-binding prediction (5 points)**
 - Appropriate HLA alleles;
 - Correct use of prediction software;
 - Proper interpretation of affinity and percentile rank.
- **Immunogenicity analysis (5 points)**

- Appropriate immunogenicity method;
- Correct score interpretation;
- Clear distinction between binding and immunogenicity.
- **Candidate prioritization and biological interpretation (5 points)**
 - Comparison with wild-type peptides;
 - Integration of mutation frequency and gene expression;
 - Well-supported selection of candidate neoantigens;
 - Discussion of limitations.

Part V. Important Rules

1. All primary output tables must be tab-delimited (.tsv).
2. Do not manually edit final tables in spreadsheet software.
3. All major steps must be reproducible from submitted code.
4. Record exact software versions and access dates for web tools.
5. Do not mix hg19/GRCh37 and hg38/GRCh38 coordinates.
6. Do not label counts, FPKM, or RPKM as "TPM".
7. Do not report peptide-MHC scores without specifying the HLA allele.
8. Do not interpret predictions as experimental validation.
9. Clearly state any assumptions.
10. Each group member must understand and explain the complete workflow.